INCH-POUND
MS51991E
w/Amendment 1
22 January 2026
SUPERSEDING
MS51991E
4 September 2025

DETAIL SPECIFICATION SHEET

# INSERT, SCREW THREAD-LOCKED IN, RING LOCKED, SERRATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-DTL-45910.

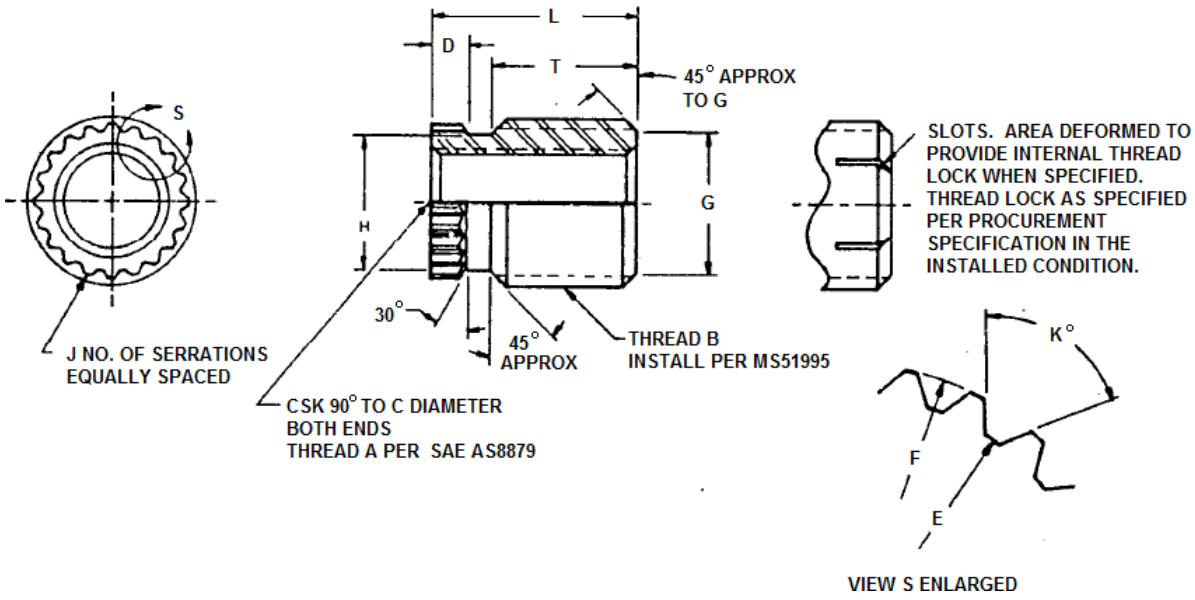



FIGURE 1. INSERT, SCREW THREAD.

Comments, suggestions, or questions on this document should be addressed to DLA Aviation-VEB, 6090 Strathmore Road, Richmond, VA 23237 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil/.

AMSC N/A FSC 5325

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TABLE I. Coarse Internal and Coarse External Threads.

| DASH NO. | A INTERNAL THREAD UNJC-3B | | B EXTERNAL THREAD SEE NOTE 4 | | | C | D | ØE | ØF | ØG | ØH | J | K° | L | T | LOCK RING PART NO. (REF) SEE NOTE 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SIZE | MINOR DIA (REF) | SIZE | PITCH DIA | MINOR DIA | | ±.015 | +.005 -.004 | +.007 -.002 | | | | +2° -1° | ±.015 | ±.015 | |
| -102 | .1120-40 | .0942<br>.0877 | .1900-24 | .1648<br>.1633 | .1407<br>.1334 | .122<br>.117 | .060 | .152 | .175 | .138<br>.133 | .148 | 14 | 102° | .250 | .150 | MS51990-102P |
| -103 | .1380-32 | .1157<br>.1076 | .2160-24 | .1908<br>.1893 | .1667<br>.1594 | .149<br>.144 | .080 | .178 | .201 | .164<br>.159 | .175 | 16 | 90° | .310 | .190 | MS51990-103P |
| -104 | .1640-32 | .1417<br>.1336 | .2500-20 | .2198<br>.2178 | .1909<br>.1824 | .175<br>.170 | .080 | .203 | .230 | .187<br>.182 | .197 | 13 | 102° | .380 | .250 | MS51990-104P |
| -105 | .1900-24 | .1600<br>.1494 | .3125-18 | .2789<br>.2769 | .2468<br>.2373 | .208<br>.198 | .090 | .255 | .284 | .237 | .248 | 17 | 86° | .440 | .290 | MS51990-105P |
| -106 | .2500-20 | .2121<br>.2013 | .3750-16 | .3372<br>.3347 | .3011<br>.2906 | .270<br>.260 | .110 | .316 | .345 | .291 | .305 | 20 | 102° | .500 | .330 | MS51990-106P |
| -107 | .3125-18 | .2690<br>.2584 | .5000-13 | .4531<br>.4506 | .4087<br>.3963 | .339<br>.324 | .120 | .380 | .407 | .396 | .370 | 24 | 102° | .560 | .360 | MS51990-107P |
| -108 | .3750-16 | .3251<br>.3142 | .5625-12 | .5116<br>.5091 | .4635<br>.4503 | .402<br>.387 | .120 | .456 | .487 | .425 | .445 | 26 | 102° | .620 | .420 | MS51990-108P |
| -109 | .4375-14 | .3795<br>.3680 | .6250-11 | .5693<br>.5668 | .5168<br>.5028 | .466<br>.451 | .140 | .567 | .601 | .503 | .553 | 26 | 111° | .690 | .460 | MS51990-109P |
| -110 | .5000-13 | .4368<br>.4251 | .7500-10 | .6885<br>.6860 | .6308<br>.6156 | .530<br>.515 | .160 | .687 | .721 | .590 | .612 | 30 | 111° | .750 | .490 | MS51990-110P |

TABLE II. Fine Internal and Coarse External Threads.

| DASH NO. | A INTERNAL THREAD UNJF-3B | | B EXTERNAL THREAD SEE NOTE 4 | | | C | D | ØE | ØF | ØG | ØH | J | K° | L | T | LOCK RING PART NO. (REF) SEE NOTE 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SIZE | MINOR DIA (REF) | SIZE | PITCH DIA | MINOR DIA | | ±.015 | +.005 -.004 | +.007 -.002 | | | | +2° -1° | ±.015 | ±.015 | |
| -202 | .1120-48 | .0971<br>.0917 | .1900-24 | .1648<br>.1633 | .1407<br>.1334 | .122<br>.117 | .060 | .152 | .175 | .138<br>.133 | .148 | 14 | 102° | .250 | .150 | MS51990-102P |
| -203 | .1380-40 | .1202<br>.1137 | .2160-24 | .1908<br>.1893 | .1667<br>.1594 | .148<br>.143 | .080 | .178 | .201 | .164<br>.159 | .175 | 16 | 90° | .310 | .190 | MS51990-103P |
| -204 | .1640-36 | .1442<br>.1370 | .2500-20 | .2198<br>.2178 | .1909<br>.1824 | .175<br>.170 | .080 | .203 | .230 | .187<br>.182 | .197 | 13 | 102° | .380 | .250 | MS51990-104P |
| -205 | .1900-32 | .1675<br>.1596 | .3125-18 | .2789<br>.2769 | .2468<br>.2373 | .206<br>.196 | .090 | .255 | .284 | .237 | .248 | 17 | 86° | .440 | .290 | MS51990-105P |
| -206 | .2500-28 | .2229<br>.2152 | .3750-16 | .3372<br>.3347 | .3011<br>.2906 | .267<br>.257 | .110 | .316 | .345 | .291 | .305 | 20 | 102° | .500 | .330 | MS51990-106P |
| -207 | .3125-24 | .2799<br>.2719 | .5000-13 | .4531<br>.4506 | .4087<br>.3963 | .336<br>.321 | .120 | .380 | .407 | .396 | .370 | 24 | 102° | .560 | .360 | MS51990-107P |
| -208 | .3750-24 | .3418<br>.3344 | .5625-12 | .5116<br>.5091 | .4635<br>.4503 | .398<br>.383 | .120 | .456 | .487 | .425 | .445 | 26 | 102° | .620 | .420 | MS51990-108P |
| -209 | .4375-20 | .3970<br>.3888 | .6250-11 | .5693<br>.5668 | .5168<br>.5028 | .462<br>.447 | .140 | .567 | .601 | .503 | .553 | 26 | 111° | .690 | .460 | MS51990-109P |
| -210 | .5000-20 | .4591<br>.4513 | .7500-10 | .6885<br>.6860 | .6308<br>.6156 | .525<br>.510 | .160 | .687 | .721 | .590 | .612 | 30 | 111° | .750 | .490 | MS51990-110P |



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TABLE III. Fine Internal and Fine External Threads.

| DASH NO. | A INTERNAL THREAD UNJF-3B | | B EXTERNAL THREAD SEE NOTE 4 | | | C | D | ØE | ØF | ØG | ØH | J | K° | L | T | LOCK RING PART NO. (REF) SEE NOTE 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | SIZE | MINOR DIA (REF) | SIZE | PITCH DIA | MINOR DIA | | ±.015 | +.005 -.004 | +.007 -.002 | | | | +2° -1° | ±.015 | ±.015 | |
| -311 | .5625-18 | .5166<br>.5084 | .8750-14 | .8316<br>.8291 | .7904<br>.7786 | .589<br>.574 | .160 | .783 | .820 | .728 | .768 | 30 | 111° | .880 | .620 | MS51990-111P |
| -312 (a) | .6250-18 | .5788<br>.5709 | 1.0000-14 | .9566<br>.9541 | .9154<br>.9036 | .685<br>.675 | .160 | .894 | .932 | .850 | .882 | 38 | 102° | 1.000 | .740 | MS51990-112P |
| -312.1 | .6250-18 | .5788<br>.5709 | 1.0000-12 | .9491<br>.9466 | .9010<br>.8878 | .685<br>.675 | .160 | .894 | .932 | .850 | .882 | 38 | 102° | 1.000 | .740 | MS51990-112P |
| -313 | .7500-16 | .6977<br>.6892 | 1.1250-12 | 1.0741<br>1.0716 | 1.0260<br>1.0128 | .777<br>.762 | .160 | 1.029 | 1.074 | .950 | 1.002 | 36 | 111° | 1.120 | .800 | MS51990-113P |
| -314 | .8750-14 | .8152<br>.8055 | 1.2500-12 | 1.1991<br>1.1966 | 1.1510<br>1.1378 | .904<br>.889 | .190 | 1.169 | 1.214 | 1.127 | 1.127 | 48 | 90° | 1.250 | .900 | MS51990-114P |
| -315 | 1.0000-12 | .9298<br>.9189 | 1.3750-12 | 1.3241<br>1.3216 | 1.2760<br>1.2628 | 1.029<br>1.014 | .190 | 1.273 | 1.318 | 1.202 | 1.252 | 42 | 111° | 1.380 | 1.020 | MS51990-115P |

(a) Inactive (not to be used for new design)

TABLE IV. Interchangeability. – see note 14

| Part Numbers | | | | | |
|---|---|---|---|---|---|
| Cancelled (CRES 17-10P) | New (CRES A-286) | Cancelled (CRES 17-10P) | New (CRES A-286) | Cancelled (CRES 17-10P) | New (CRES A-286) |
| MS51991 | MS51991 | MS51991 | MS51991 | MS51991 | MS51991 |
| C102 | E102 | C202 | E202 | C311 | E311 |
| C103 | E103 | C203 | E203 | C312 | E312 |
| C104 | E104 | C204 | E204 | C312.1 | E312.1 |
| C105 | E105 | C205 | E205 | C313 | E313 |
| C106 | E106 | C206 | E206 | C314 | E314 |
| C107 | E107 | C207 | E207 | C315 | E315 |
| C108 | E108 | C208 | E208 | | |
| C109 | E109 | C209 | E209 | | |
| C110 | E110 | C210 | E210 | | |

NOTES:

1. Material:
   - Steel, carbon, composition C117 in accordance with ASTM A108
   - Steel, corrosion-resistant, Type A286 (UNS S66286) per SAE AMS5731, SAE AMS5732, SAE AMS5734 or SAE AMS5737

2. Protective coating:
   - Steel, carbon shall be cadmium plated in accordance with SAE AMS-QQ-P-416, Type II, Class 3
   - Steel, carbon shall be zinc-nickel plated in accordance with MIL-PRF-32647, Type K16
   - Steel, corrosion-resistant, shall be passivated in accordance with SAE AMS2700
   - Steel, corrosion-resistant, shall be passivated in accordance with SAE AMS2700, Method 1, Type 6, Type 7, or Type 8, no post treatment, or Method 2, no post treatment
   - Steel, corrosion-resistant, with internal thread lock for aerospace applications shall also be solid film lubricant coated per SAE AS5272 Type I[1]
   - Steel, corrosion-resistant, with internal thread lock for non-aerospace applications shall also be solid film lubricant coated per MIL-PRF-46010[2]

[1]SAE AS5272 Type 1 lubricant is technically equivalent to MIL-L-46010 Type I lubricant which superseded MIL-L-8937 lubricant used in previous revisions.

[2]MIL-PRF-46010 lubricant is lead (Pb) free and is not technically equivalent to MIL-L-46010 Type I lubricant which superseded MIL-L-8937 lubricant used in previous revisions. Use of MIL-PRF-46010 in aerospace applications must first be validated.



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3. Surface roughness: Machine surfaces shall be 125 microinches in accordance with ASME B46.1 except serrated collar.

4. Threads: The external thread has a special pitch diameter and minor diameter which installs into a national class 3 tapped hole. Threads also shall be in accordance with procurement specification.

5. Heat treatment: Inserts shall be capable of developing 125,000 PSI tensile strength as specified in the procurement specification.

6. Hardness: – Steel, carbon, Rockwell B 78 minimum
– Steel, corrosion-resistant, Rockwell B 83 minimum

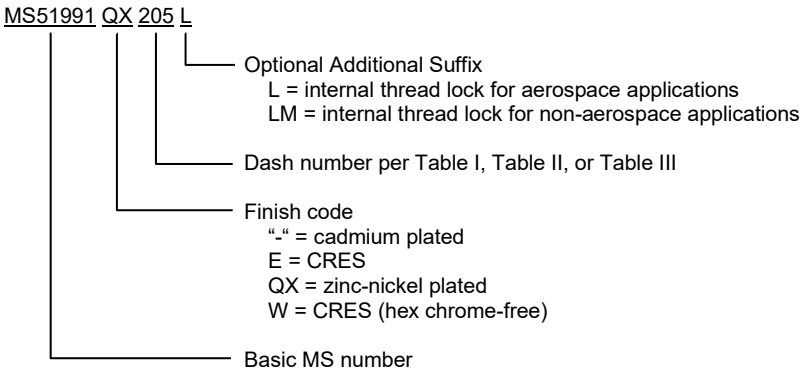
7. Fillets: .015 R max.

8. Edges: Break sharp edges .003 - .015 unless otherwise specified.

9. Dimensions: Dimensions are in inches and are met after plating.

10. Tolerances: Linear dimensions ±.005, angular dimensions ±2°.

11. Part numbers:




| | | |
|---|---|---|
| Examples: | MS51991-205 | Insert, carbon steel, non-locking, cadmium plated |
| | MS51991QX205 | Insert, carbon steel, non-locking, zinc-nickel plated |
| | MS51991E205 | Insert, CRES, non-locking, passivated |
| | MS51991W205 | Insert, CRES, non-locking, hex chromium-free, passivated |
| | MS51991-205L | Insert, carbon steel, internal thread lock, aerospace applications. |
| | MS51991QX205L | Insert, Insert, carbon steel, internal thread lock, aerospace applications, zinc-nickel plated |
| | MS51991E205L | Insert, CRES, internal thread lock, aerospace applications |
| | MS51991E205LM | Insert, CRES, internal thread lock, non-aerospace applications |
| | MS51991W205LM | Insert, CRES, internal thread lock, non-aerospace applications, hex chrome-free passivated |

12. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

13. Referenced documents shall be of the same issue in effect on date of invitation for bid.

14. The dash numbers "C100, C200, and C300" series CRES inserts in the original issue of this standard are cancelled/inactivated after approval date of revision "A" indicated on this document. The cancelled inserts should be used on existing callouts until stock is depleted. Use the new "E100, E200, and E300" series CRES inserts for replacement of "C100, C200, and C300" series CRES inserts in accordance with Table IV.

15. For applicable locking dash number coding see SAE AS51990.

16. Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue



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APPENDIX A

3D Model of MS51991-207 for reference only.
Viewing this model requires a PDF viewer that supports 3D models.
Adobe Acrobat is known to support 3D models but there may be other suitable viewers.



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MILITARY INTEREST

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Preparing Activity
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(Project 5325-2026-001)

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